

Detailed Notes on Human-Computer Interaction (HCI)

Fall 2025

Abstract

These notes provide a comprehensive overview of the introductory concepts of Human-Computer Interaction (HCI). The document covers the course's administrative details, fundamental definitions of interaction design, the importance of good and bad design principles through various examples, and the goals of usability and user experience. Furthermore, it delves into the interdisciplinary nature of HCI, the process of designing for diverse user needs, including accessibility and cultural differences, and key design principles such as visibility, feedback, constraints, consistency, and affordances.

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1 Introduction and Housekeeping (Slides 1-2)

1.1 Course Overview

The course, **Human-Computer Interaction (HCI)**, for Fall 2025, aims to introduce the fundamental principles and practices of designing effective and intuitive interactive systems.

1.2 Administrative Details

- **Lecture Hours:** Mondays & Wednesdays, 8:30 am – 9:45 am.
- **Tutorials:** Attendance in one tutorial slot during the week is mandatory.
- **Counselling Hours:** Mondays & Wednesdays, 10:00 am – 11:30 am.
- **Contact Information:**
 - Shakeel Khoja: skhoja@iba.edu.pk
 - Afshan Ejaz: aejaz@iba.edu.pk
 - The Course Learning Management System (LMS) will also be a primary mode of contact.
- **Course Rules:**
 - Active participation through asking questions is encouraged.
 - No extensions will be granted for deadlines.
 - Academic integrity is paramount; no plagiarism will be tolerated.
 - All group members are expected to contribute equally (no free riding).

2 Examples of Interactive Designs (Slides 3-4)

These slides present visual examples of interactive technologies to stimulate thinking about how users interact with different devices.

- **Game Boy Advance SP:** A classic example of a portable gaming device with physical buttons, a D-pad, and a clamshell design. The interaction is direct and tactile.
- **Project Cafe Concept Controller (Nintendo Wii U):** This showcases an evolution in gaming interaction, combining a traditional controller layout with an integrated touchscreen. This design enables "second-screen" experiences, where the controller can display supplementary information or even a different viewpoint of the game.
- **Motion-Sensing Games (e.g., Xbox Kinect):** The image shows children interacting with a large screen display by moving their bodies. This represents a shift from controller-based interaction to more natural, gesture-based user interfaces. The Chinese text on the screen, “” (Games change life), emphasizes the transformative impact of such interactive experiences.

3 What’s in a Name? Perspectives on Interaction Design (Slide 5)

Interaction Design (ID) is a broad field, and different terms are used to describe it, each emphasizing a slightly different perspective.

- **User-Centered Design (UCD):** This approach places the user at the center of the design process. It focuses on the user’s tasks, goals, and workflow. For example, when designing an e-commerce website, a UCD approach would involve researching how users typically shop online and designing the checkout process to be as simple and intuitive as possible for them.
- **Human-Centered Design (HCD):** A broader term than UCD, HCD considers the human aspect in its entirety, including cognitive, physical, and emotional factors. It is not limited to just the "user" of a product but considers the wider human context.

- **People-Centered Design:** This term is often used to be more inclusive and less technical than "user." It emphasizes designing for real people in their everyday lives.
- **Customer Experience (CX) Design:** This perspective is more business-oriented and focuses on the entire journey a customer has with a company or brand, of which the interaction with a specific product is only one part.

The slide poses a question: "Which do you think is preferable and why?" There is no single correct answer. **Human-centered design** is often preferred in academic and ethical contexts because it is the most encompassing and emphasizes the holistic well-being of the individual beyond their role as a mere "user" or "customer."

4 Good and Bad Design (Slides 6-11)

This section contrasts well-designed products with poorly designed ones to illustrate core HCI principles.

4.1 Good Design: The TiVo Remote (Slide 6)

The TiVo remote is presented as a classic example of good design.

- **Ergonomics:** Its "peanut shape" is designed to fit comfortably in the human hand.
- **Logical Layout and Visual Cues:** Buttons are grouped by function (e.g., navigation, playback). Color-coding (like the red Netflix button) and distinctive shapes (the thumbs up/down buttons) help users quickly identify and locate functions.
- **Discoverability:** The layout is intuitive, making it easy for users to find the buttons they need without looking.

A link to an article on Gizmodo is provided for further reading: <http://gizmodo.com/5017972/story-of-a-peanut-the>

4.2 Bad Designs

4.2.1 Elevator Controls (Slide 7)

The example shows an elevator control panel where the labels and the buttons in the bottom row are visually identical. This lack of differentiation makes it easy for a user to mistakenly press a label instead of the intended button. The buttons on the top row (for floor numbers) do not have this issue because the numbers themselves act as clear, distinct labels. This illustrates a failure in providing clear **signifiers**.

4.2.2 Vending Machine (Slide 8)

This vending machine exhibits poor design by violating established user conventions.

- **Unconventional Workflow:** It requires the user to first press a button to activate the bill reader, which is counter-intuitive.
- **Violation of Convention:** Most users expect to insert payment first and then make their selection. By reversing this order, the design forces users to learn a new, unnecessary process, leading to confusion and frustration.

4.2.3 Ambiguous Road Sign (Slide 9)

The image displays a road sign with a "Do Not Enter" symbol next to a right-pointing arrow. This sends a contradictory message: "Do not go this way" and "Go this way." Clear communication is a fundamental aspect of good design, and this sign fails to provide it.

4.2.4 Confusing Remote Control (Slide 10)

A remote control is shown where the volume and channel buttons are represented by identical up/down triangle symbols, differentiated only by tiny letters ("V" and "CH"). This makes them difficult to distinguish at a glance, violating the principle of clear visual differentiation.

4.2.5 Bad Usability Calendars (Slide 11)

These calendars from badusability.com highlight common design flaws in information display:

- **January:** The graphic of the Titanic obstructs the dates, illustrating how decorative elements can interfere with primary functionality.
- **February:** The layout is chaotic, with days of the week not in a standard grid format. The week numbers are present, but the overall structure is confusing.
- **November:** This example is filled with annotations pointing out flaws like missing dates, inconsistent labeling (single vs. double letters for days), and illogical placement of the week number.
- **December:** This calendar is more conventional, but still has minor issues that could be improved for better clarity.

5 Information Overload and Outdated Models (Slides 12-13)

5.1 Too Much Data (Slide 12)

The image of a Pakistani passport is used to illustrate the concept of information overload. The passport contains a large amount of data, some of which is redundant (e.g., information is present in both human-readable and machine-readable formats). In design, it's crucial to present only the necessary information in a clear and organized manner to avoid overwhelming the user.

5.2 Outdated Models (Slide 13)

This slide shows a 1954 RAND Corporation model of a "home computer." The accompanying text reveals the predictions made at the time, which were both prescient (the idea of a home computer) and flawed (the belief that it would require "not yet invented technology" and use a "teletype interface and the Fortran language"). This serves as a reminder that our current design paradigms are not permanent and will evolve. What seems advanced today may be considered outdated tomorrow.

6 Health and Accessibility (Slides 14-18)

This section uses Ishihara plates, which are used to test for color blindness, to metaphorically and literally introduce the topic of designing for health and accessibility. The ability to perceive the numbers within the plates (7, 2, 45, 74) depends on the individual's color vision. This highlights a critical point: designers cannot assume that all users perceive the world in the same way. Design choices, especially regarding color, can create barriers for users with disabilities like color blindness.

7 Good Design Revisited and Its Benefits (Slides 19-20)

7.1 The Marble Answering Machine (Slide 19)

This conceptual design by Durrell Bishop (1995) is a prime example of good interaction design.

- **How it works:** For each incoming message, a marble is dispensed into a tray. To listen to a message, the user places a marble into a small indentation. To call the person back, the user places that marble into another slot.
- **Intuitive Design:** It leverages our understanding of how everyday physical objects behave. The one-to-one mapping between marbles and messages is simple and clear.
- **Pleasurable Experience:** The design is not just functional but also aesthetically pleasing and engaging to use.
- **Simplicity:** Core tasks require only single, straightforward actions.

7.2 Benefits of Good Design (Slide 20)

Good design seamlessly integrates into our daily lives, making complex tasks feel simple.

- Withdrawing cash from an ATM.
- Becoming engrossed in a computer game.
- Using a spreadsheet.
- Shopping online.
- Using social media on a mobile phone.

In all these examples, good design reduces cognitive load and allows users to focus on their goals, not on figuring out the interface.

8 Dilemma: Interacting with Smart TVs (Slides 22-23)

Smart TVs present a significant design challenge: how to enable complex interactions, like text entry for searching, from a distance (the "10-foot experience").

- **Standard Remote / Grid Keyboard:** Pecking at an on-screen keyboard with a D-pad is slow and cumbersome.
- **Touchpad Remote:** Swiping across alphanumeric rows (like on the Apple TV remote) can be faster but may lack precision.
- **Voice Control:** Using a remote with a built-in microphone or a smart speaker is often the most efficient method for search queries, though it can be prone to errors in noisy environments or with complex terms.
- **Physical Keyboard (Minuum's Keyboard):** While potentially fast, a full physical keyboard is often considered too bulky for a living room environment.

This dilemma highlights that there is often no single "best" solution. The optimal choice depends on the context, the user's preferences, and the specific task at hand.

9 The Shift to Digital (Slide 24)

Many activities that once required physical artifacts are now digital.

- **Examples:** Paying bills, hailing a taxi, using maps, and ticketing for travel.
- **Benefits:**
 - **Convenience:** Tasks are often faster and easier.
 - **Efficiency:** Information can be stored and reused (e.g., saved payment details).
 - **Accessibility:** Services can be accessed from anywhere, without waiting in line.
- **Disadvantages:**
 - **Security Risks:** Concerns about fraud and data privacy.
 - **Digital Divide:** Not everyone has equal access to or comfort with digital tools.
 - **Loss of Human Interaction:** Some people prefer face-to-face communication, especially for important transactions.

10 What to Design: Core Considerations (Slide 25)

To design effectively, one must understand the context of use.

- **Who are the users?** (e.g., children, experts, elderly)
- **What activities are they performing?** (e.g., browsing, creating, learning)
- **Where is the interaction taking place?** (e.g., at a desk, on the go, in a public space)

The goal is to optimize the interaction to match the users' activities and needs within their specific context.

11 Defining Interaction Design (ID) (Slide 26)

Two key definitions are provided:

- *"Designing interactive products to support the way people communicate and interact in their everyday and working lives."* (Sharp, Rogers, and Preece, 2019) - This definition focuses on the purpose of ID: to support human activities.
- *"The design of spaces for human communication and interaction."* (Winograd, 1997) - This broader definition highlights that ID is not just about screens but about creating environments (both physical and digital) where interaction occurs.

12 Goals of Interaction Design (Slide 27)

The overarching goals of ID are to create products that are both usable and provide a high-quality user experience.

- **Develop usable products:** The product should be effective, efficient, and satisfying to use.
- **Involve people in the design process:** This is the core of user-centered design.
- **Consider human capabilities:** Design should leverage what people are good at and provide support for tasks they find difficult.
- **Address user needs and privacy:** Understand existing workflows and be mindful of data collection.
- **Use people-centered techniques:** Employ methods like user research, prototyping, and usability testing throughout the design process.

13 The Interdisciplinary Nature of HCI (Slides 28-34)

13.1 ID as an Umbrella Term (Slide 28)

Interaction Design is a broad field that encompasses many other disciplines, including:

- User Interface (UI) Design
- User Experience (UX) Design
- Software Design
- Product Design
- Web Design

13.2 Contributing Fields (Slides 29-32)

HCI and ID draw upon a wide range of fields, which can be grouped into three categories:

1. Academic Disciplines:

- **Psychology/Cognitive Science:** Provides understanding of human perception, memory, and problem-solving.
- **Social Sciences (Sociology, Anthropology):** Offer insights into how people interact in groups and within different cultural contexts.
- **Computer Science/Engineering:** Provides the technical foundation for building interactive systems.
- **Ergonomics/Human Factors:** Focuses on the physical fit between people and products.
- **Informatics:** Deals with the structure, behavior, and interactions of natural and artificial computational systems.

2. Design Practices:

- Graphic Design, Product Design, Industrial Design, Film Industry.

3. Interdisciplinary Fields:

- HCI, Ubiquitous Computing (UbiComp), Computer-Supported Cooperative Work (CSCW).

13.3 Working in Multidisciplinary Teams (Slide 33)

- **Benefits:** Bringing together people with different backgrounds and perspectives leads to more creative and robust design solutions.
- **Disadvantages:** Differences in terminology, priorities, and methodologies can create communication challenges that need to be managed.

13.4 Interaction Design in Business (Slide 34)

Many successful consultancies specialize in ID and user experience.

- **Nielsen Norman Group:** A leading voice in usability research, focusing on human-centered design.
- **Cooper:** Known for pioneering goal-directed design and the use of personas.
- **IDEO:** A world-renowned design and innovation firm that champions design thinking.

14 User Experience (UX) (Slides 35-37)

14.1 People-Centered Design (Slide 35)

This approach goes beyond just functionality and usability to consider the emotional aspect of interaction.

- It involves understanding how a product makes a person feel.
- As Don Norman (2004) states, it's not enough for products to be functional and usable; they should also bring "joy and excitement, pleasure and fun, and yes, beauty to people's lives."

14.2 Defining the User Experience (Slide 36)

UX encompasses a user's perceptions and feelings before, during, and after their interaction with a product or service.

- **Hassenzahl's (2010) Model:**
 - **Pragmatic Qualities:** Relate to usability—how simple and practical the product is for achieving a goal.
 - **Hedonic Qualities:** Relate to the pleasure and stimulation derived from using the product. This has become increasingly important, especially in relation to user well-being.

14.3 Example of Great UX: The iPod (Slide 37)

The success of the iPod was not just due to its technical specifications. It was a result of a meticulously crafted user experience.

- **Holistic Experience:** From the unboxing to the "click wheel" interface and the integration with the iTunes software, every touchpoint was designed to be simple, elegant, and pleasurable.
- **Emotional Appeal:** It was positioned as a cool, must-have fashion item, creating a strong and distinct brand identity.

15 Core Characteristics and Design Considerations (Slides 38-44)

15.1 Core Characteristics of ID (Slide 38)

1. Users should be involved throughout the entire design and development process.
2. Usability and user experience goals should be explicitly defined and agreed upon at the start of a project.
3. Iteration is essential. The process of design, prototyping, and evaluation should be repeated to refine the product.

15.2 Why These Characteristics Are Important (Slide 39)

Following these principles helps designers:

- Understand what users truly want and need.
- Recognize that "one size does not fit all" and design for diverse user groups.
- Avoid designing based on incorrect assumptions.
- Be sensitive to users' capabilities and limitations.

15.3 Accessibility and Inclusiveness (Slide 40)

- **Accessibility:** The extent to which a product can be used by people with disabilities. This often involves designing for assistive technologies like screen readers (e.g., Apple VoiceOver).
- **Inclusiveness:** A broader concept of making products that accommodate the widest possible range of people, regardless of disability, age, education, or income. Smartphones are a good example, as they are designed to be used by a vast and diverse global population.

15.4 Understanding Disability (Slides 41-43)

- Disability is not a fixed state; it can change over time.
- Impairments can be classified as sensory, physical, or cognitive.
- Impairments can be categorized by duration:
 - **Permanent:** A long-term condition.
 - **Temporary:** Caused by an accident or illness.
 - **Situational:** Caused by the environment (e.g., not being able to hear in a noisy room).
- The final slide in this sub-section shows fashionable prosthetic leg covers, illustrating a modern approach to disability that focuses on desirability and self-expression, not just functionality.

15.5 Cultural Differences (Slide 44)

Designers must be aware of cultural differences that can impact how a product is perceived and used.

- **Date Formats:** The ambiguity of ‘5/21/2023’ (MM/DD/YYYY in the US) vs. ‘21/5/2023’ (DD/MM/YYYY in many other parts of the world) highlights the need for clear standards (like ISO 8601: YYYY-MM-DD) in international products.
- **Universal vs. Localized Design:** While some products like smartphones have achieved near-universal acceptance, other products, particularly websites, often require significant localization of content, layout, and imagery to be effective in different cultures.

16 Usability and User Experience Goals (Slides 45-47)

16.1 Usability Goals (Slide 45)

These are objective criteria that can be measured to assess how usable a product is.

- **Effective to use:** Can users achieve their goals?
- **Efficient to use:** How much effort does it take?
- **Safe to use:** Does it prevent users from making serious errors?
- **Have good utility:** Does it have the necessary functionality?
- **Easy to learn:** How quickly can a new user become proficient?
- **Easy to remember:** Can a returning user pick it up again easily?
- **Enjoyable to use:** This goal bridges usability and user experience.

16.2 Usability vs. User Experience Goals (Slide 46)

- UX goals are concerned with the subjective feelings and emotions of the user, which are harder to measure than objective usability metrics.
- There can be trade-offs. For example, a highly secure system (good usability) might require many tedious steps, making it less enjoyable (poor UX). A fun game (good UX) might not prioritize safety in the same way a banking application would.

16.3 User Experience Goals (Slide 47)

This slide lists a range of desirable and undesirable terms that can be used to describe the user experience.

- **Desirable Aspects:** Satisfying, enjoyable, fun, engaging, helpful, motivating.
- **Undesirable Aspects:** Boring, frustrating, patronizing, creepy, intrusive, annoying.

These terms provide a vocabulary for designers to articulate and aim for specific emotional outcomes.

17 Design Principles (Slides 48-62)

Design principles are high-level guidelines that can help designers make informed decisions.

17.1 Visibility (Slides 49-51)

This principle, popularized by Don Norman, states that the relevant parts of a system should be visible, and it should be obvious what they do.

- **Poor Visibility Example:** An elevator panel where the card slot required for operation is not apparent. The user doesn't know what to do because the necessary control is hidden.
- **Improving Visibility:** Making the card reader more obvious, adding a label, or using a flashing light to draw attention to it.
- **Invisible Controls:** The automatic faucet that doesn't work with black clothing is another example where the underlying mechanism (an infrared sensor) is invisible and has a hidden constraint, leading to a frustrating user experience.

17.2 Feedback (Slide 52)

Feedback is about sending information back to the user to let them know what has happened.

- **Forms of Feedback:** Sound, highlighting, animation, vibration.
- **Example:** When a button on a screen is clicked, it might make a "click" sound or change its appearance (e.g., become highlighted or appear pressed down) to confirm that the input was received.

17.3 Constraints (Slide 53)

Constraints are used to restrict the kinds of actions that can be performed, which helps to prevent errors.

- **Physical Constraint Example:** A key that can only be inserted into a lock one way.
- **Digital Constraint Example:** "Graying out" a "Save" menu option when no changes have been made to a document. This constrains the user from performing an irrelevant action.

17.4 Logical Design and Mapping (Slides 54-55)

This principle relates to having a clear relationship between controls and their effects.

- **Ambiguous Design:** The back of a PC where the mouse and keyboard ports are not clearly marked, leading to confusion.
- **Logical Design Solutions:**
 - (A) **Adjacency:** Placing the icon directly next to the corresponding connector.
 - (B) **Color-Coding:** Using color to visually link the port with the label.

17.5 Consistency (Slides 56-59)

Consistency means designing interfaces to have similar elements for similar tasks.

- **Benefit:** Consistent interfaces are easier to learn and use. For example, using 'Ctrl+C' for copy and 'Ctrl+V' for paste across different applications.
- **Internal Consistency:** Consistency within a single application.
- **External Consistency:** Consistency across different applications and devices.
- **Inconsistency Example (Slide 59):** The number layout on a telephone keypad (1-2-3 on the top row) is different from the layout on a calculator or computer number pad (7-8-9 on the top row). This is a well-known case of external inconsistency that forces users to switch mental models.

17.6 Affordances (Slides 60-62)

An affordance is a property of an object that suggests how it can be used.

- **Physical Affordance:** A door handle "affords" pulling or turning. A button "affords" pushing.
- **Perceived Affordances in Digital Interfaces:** Since interface elements are virtual, they don't have real affordances. Instead, they have "perceived" affordances, which are learned conventions. For example, we have learned that an icon that looks like a button with a shadow effect can be clicked.
- **Activity (Slide 62):** The slide shows various screen objects and asks the user to consider what actions they afford. A novice user might not know what to do with them, highlighting that perceived affordances rely on learned conventions.

18 Key Points Summary (Slide 63)

- Interaction design is about creating interactive products to support people's lives.
- The goal is to create high-quality user experiences.
- ID is a multidisciplinary field.
- Effective design requires considering many factors, including context, user groups, UX goals, accessibility, and cultural differences.
- Design principles (like visibility, feedback, and consistency) act as useful heuristics for designing and evaluating interactive products.